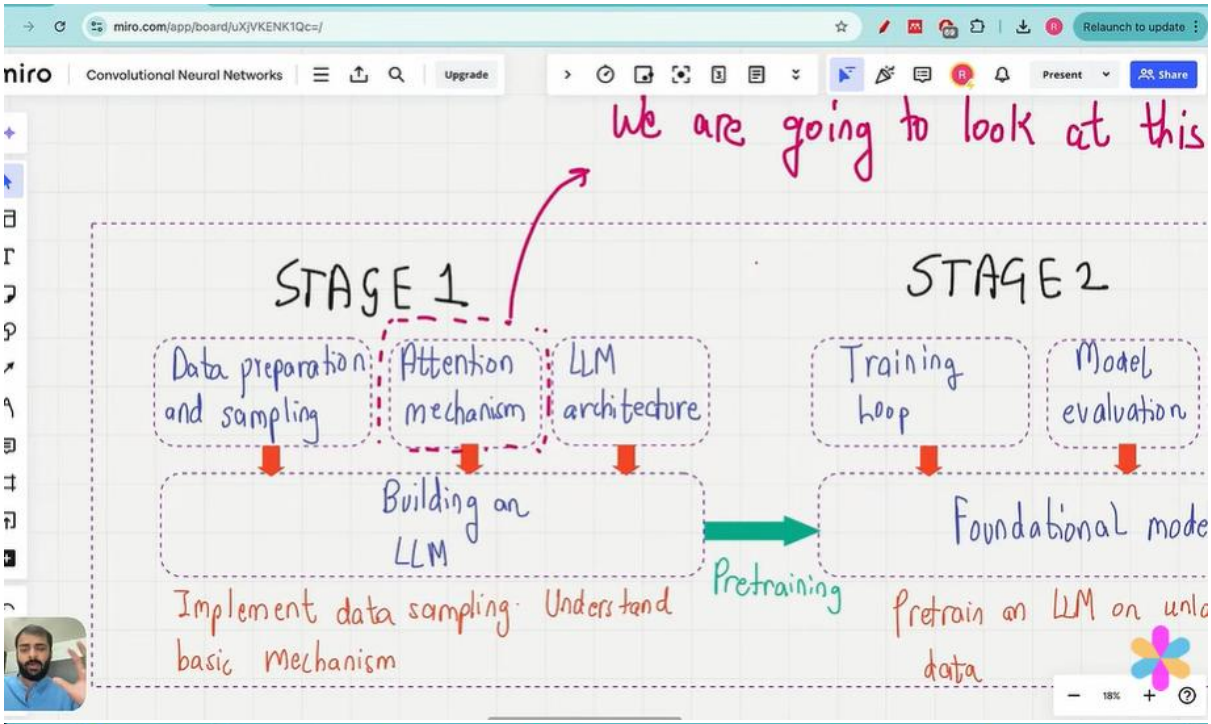
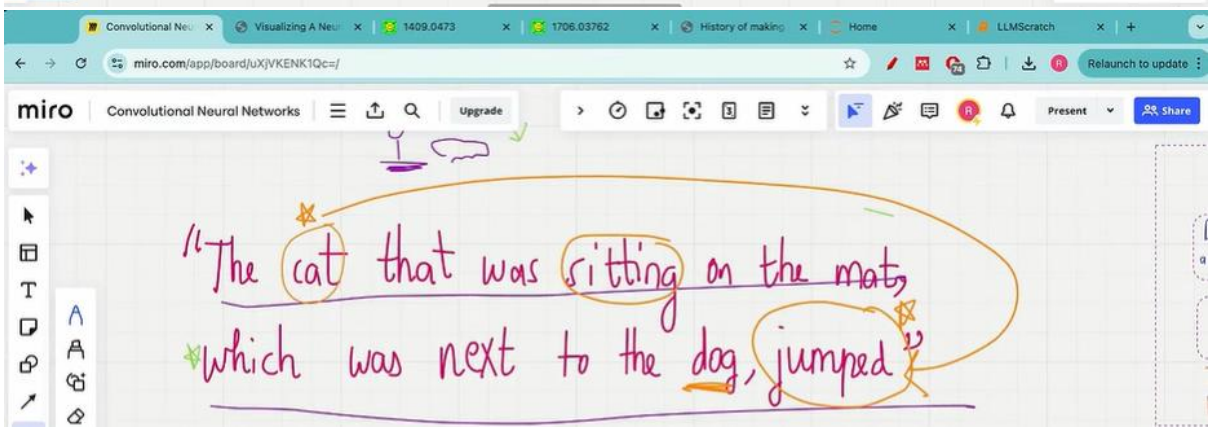
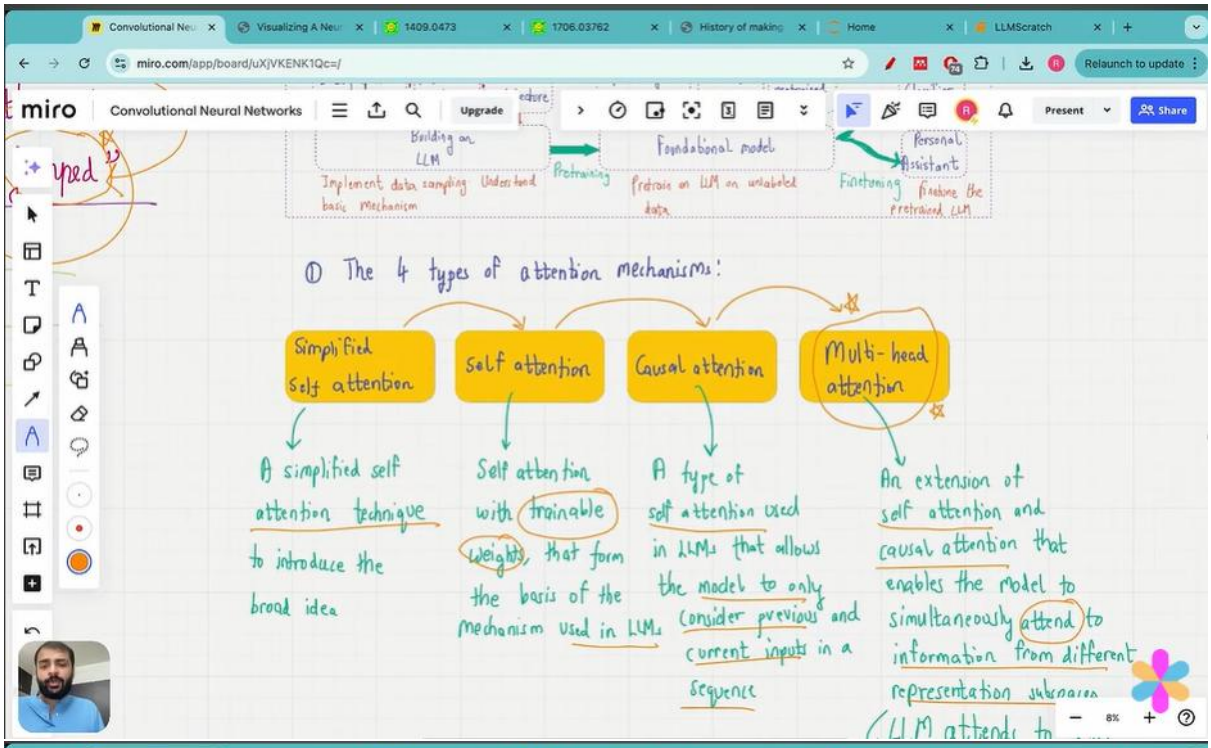
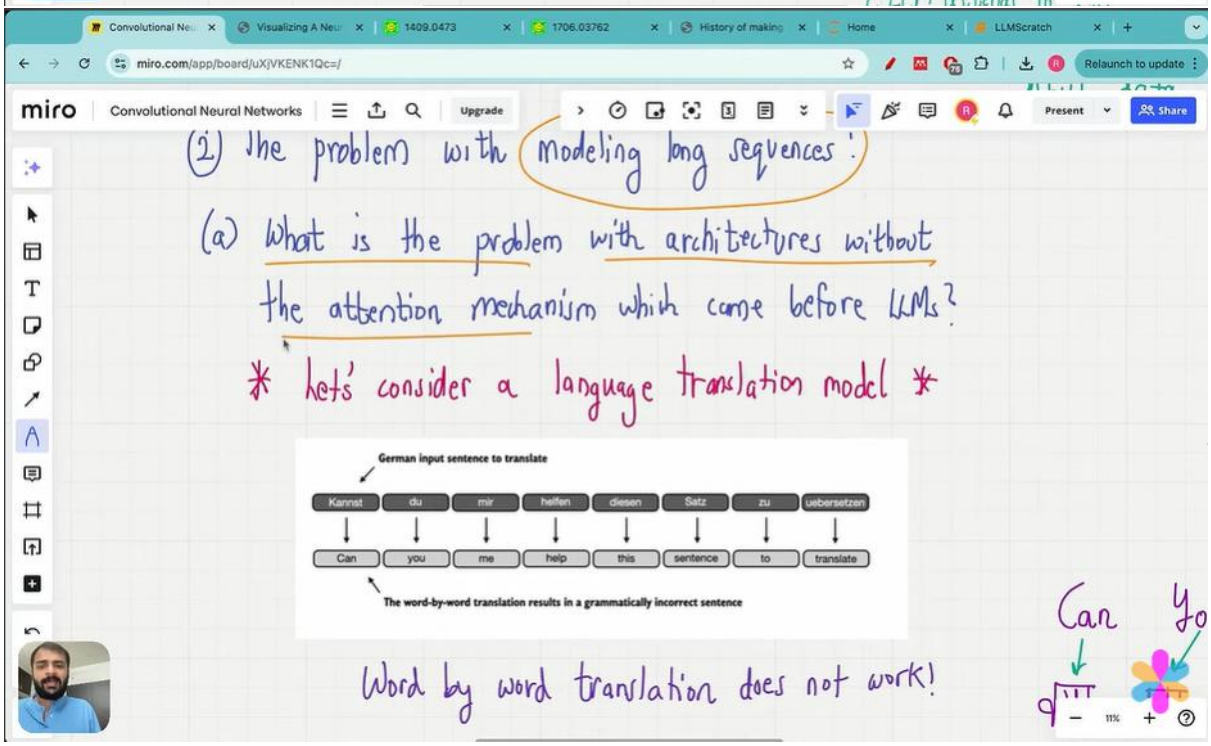
Lec 13


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Convolutional Neural Networks
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Present
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We are going to look at this
STAGE 1
STAGE 2
Data preparation and sampling
Attention mechanism
LLM architecture
Training loop
Model evaluation
Building an LLM
Foundational mode
Pretraining
Implement data sampling
Understand basic mechanism
Pretrain an LLM on unla
data


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"The cat that was sitting on the mat,
which was next to the dog, jumped"


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Building an LLM
Implement data sampling
Understand basic mechanism
Pretraining
Foundational model
Pretrain an LLM on unlabeled data
Finetuning
Personal Assistant
finetune the pretrained LLM
① The 4 types of attention mechanisms:
Simplified self attention
Self attention
Causal attention
Multi-head attention
A simplified self attention technique to introduce the broad idea
Self attention with trainable weights that form the basis of the mechanism used in LLMs
A type of self attention used in LLMs that allows the model to only consider previous and current inputs in a sequence
An extension of self attention and causal attention that enables the model to simultaneously attend to information from different representation subspaces
LLM attends to


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(2) The problem with modeling long sequences:
(a) What is the problem with architectures without the attention mechanism which came before LLMs?
* let's consider a language translation model *
German input sentence to translate
Kannst du mir helfen diesen Satz zu uebersetzen
Can you me help this sentence to translate
The word-by-word translation results in a grammatically incorrect sentence
Word by word translation does not work!
Can
Yo

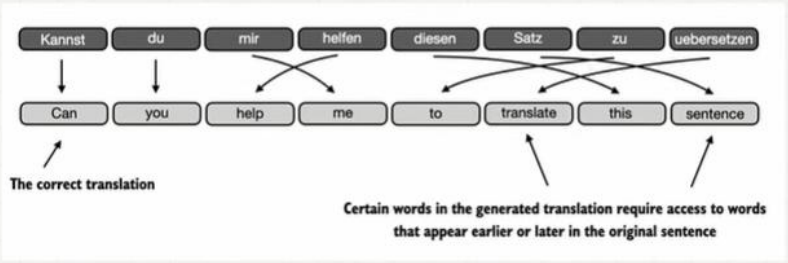
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Word by word translation does not work!



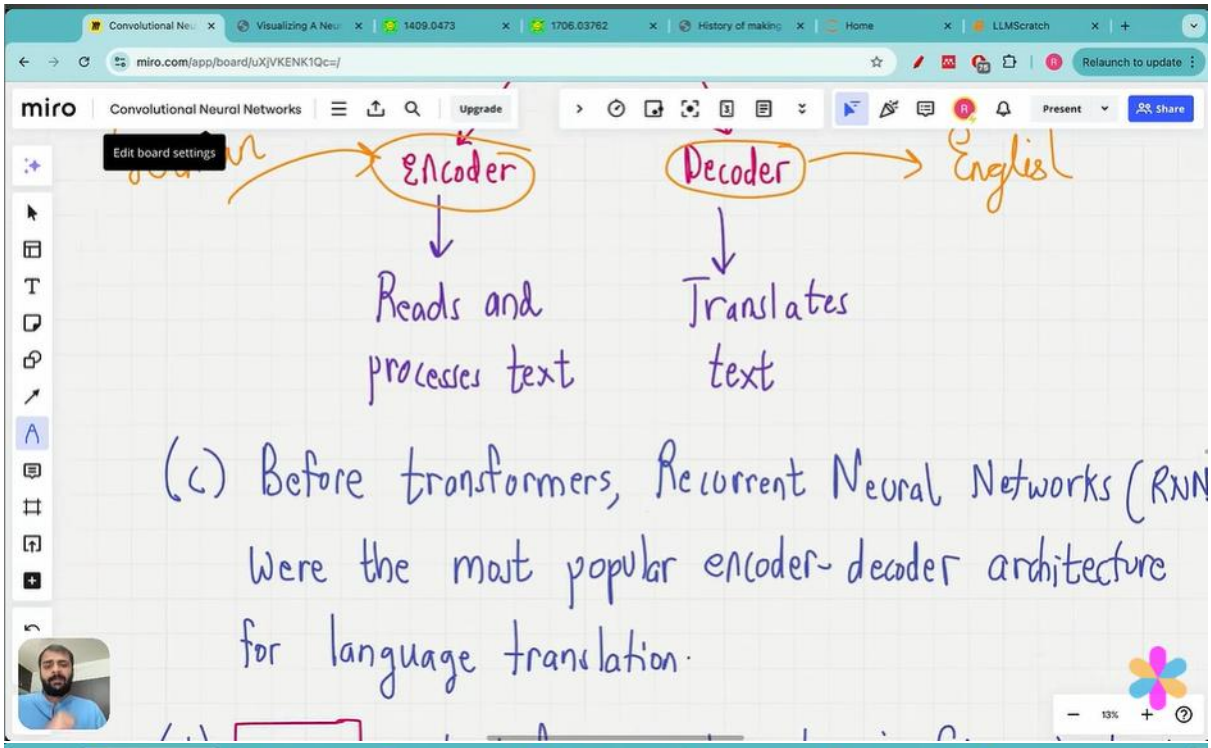
The translation process requires (contextual) understanding and grammar alignment.

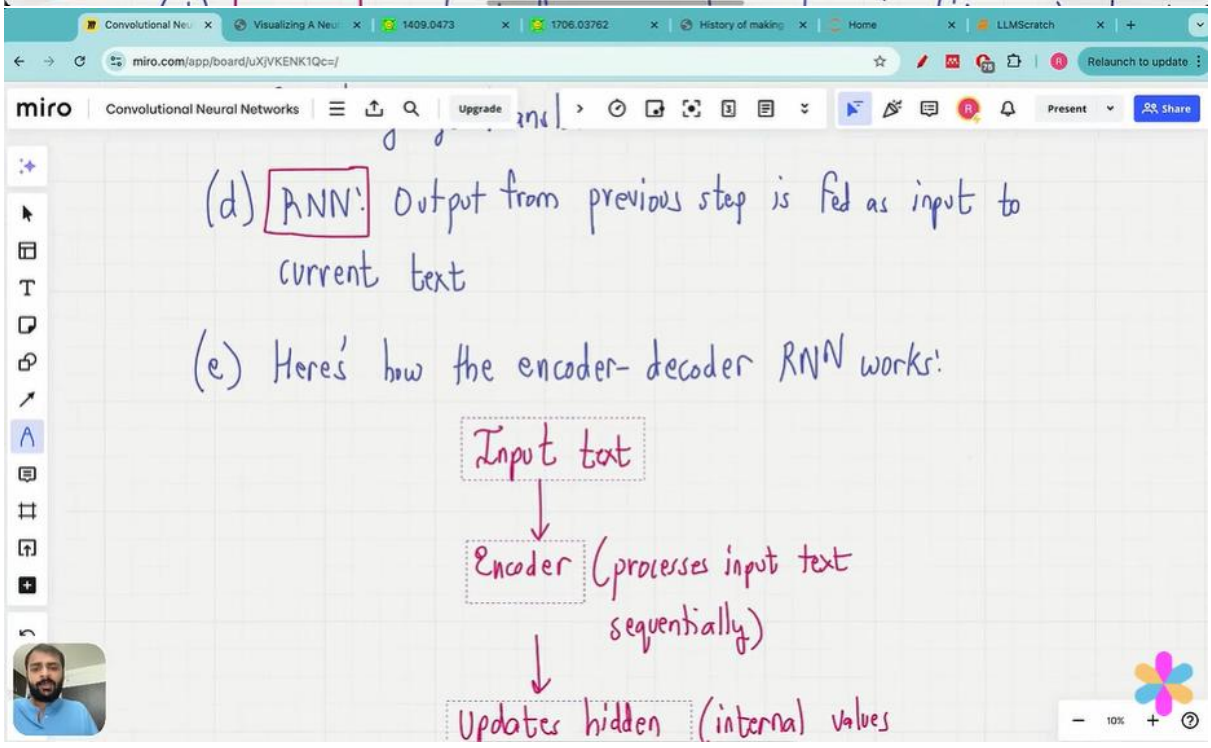
The translation process requires (contextual) understanding and grammar alignment.

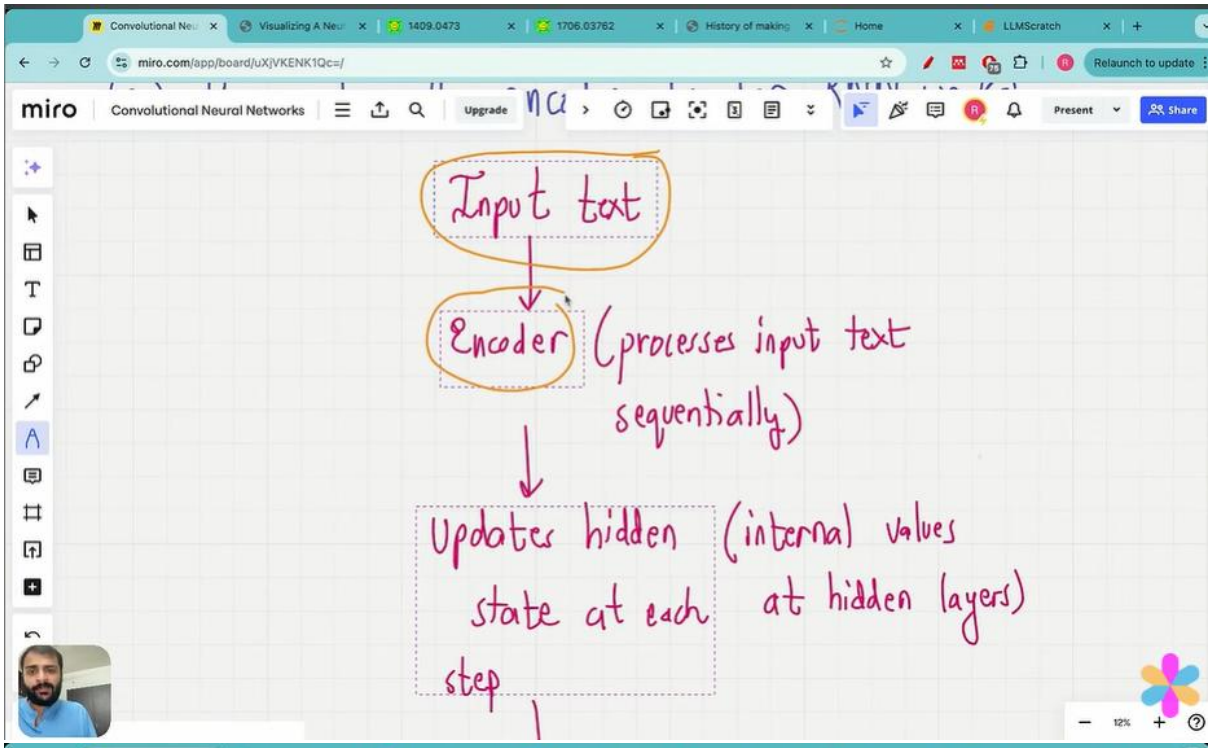
(b) To address this issue that we cannot translate text word by word, it is common to use a neural network with two submodules:

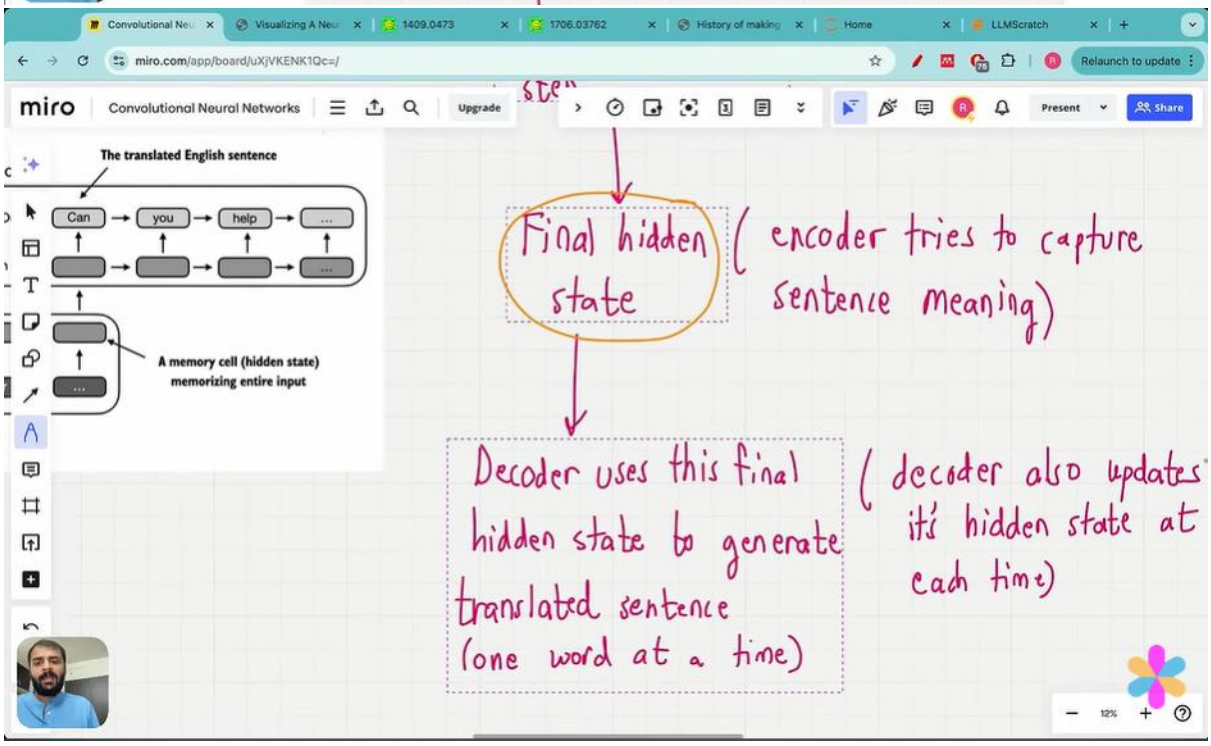
Encoder → Reads and

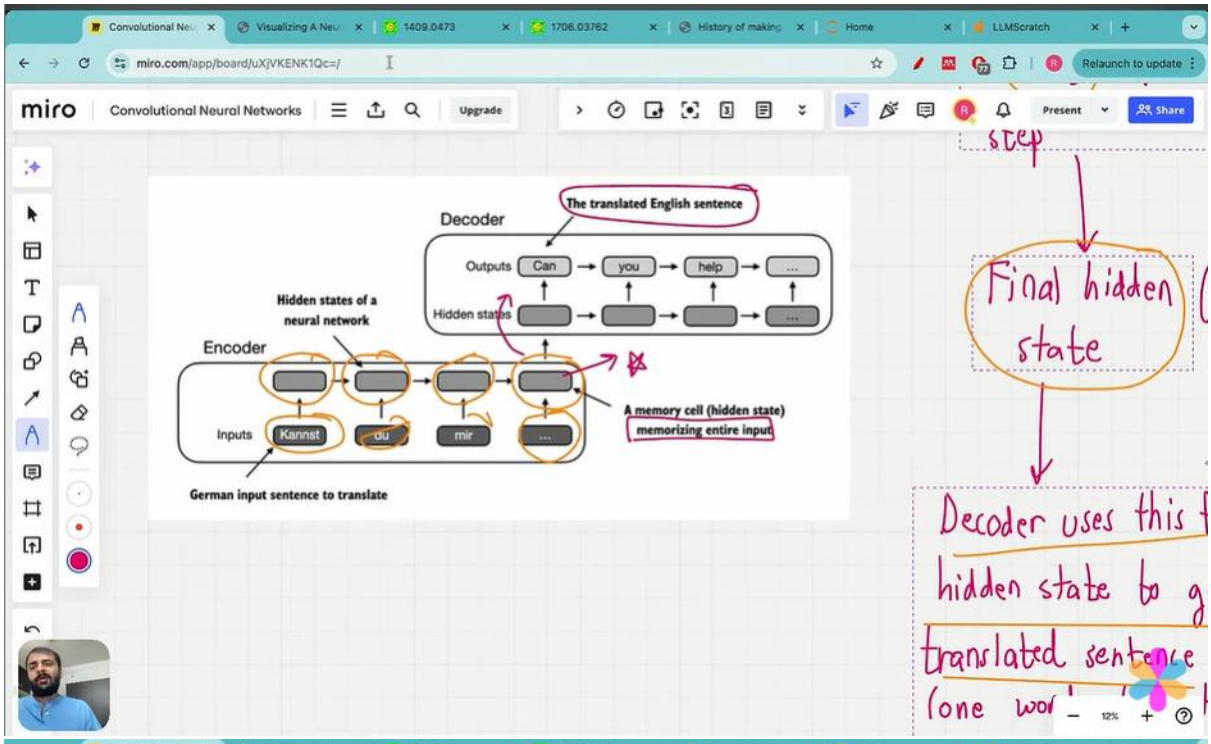
Decoder → Translates

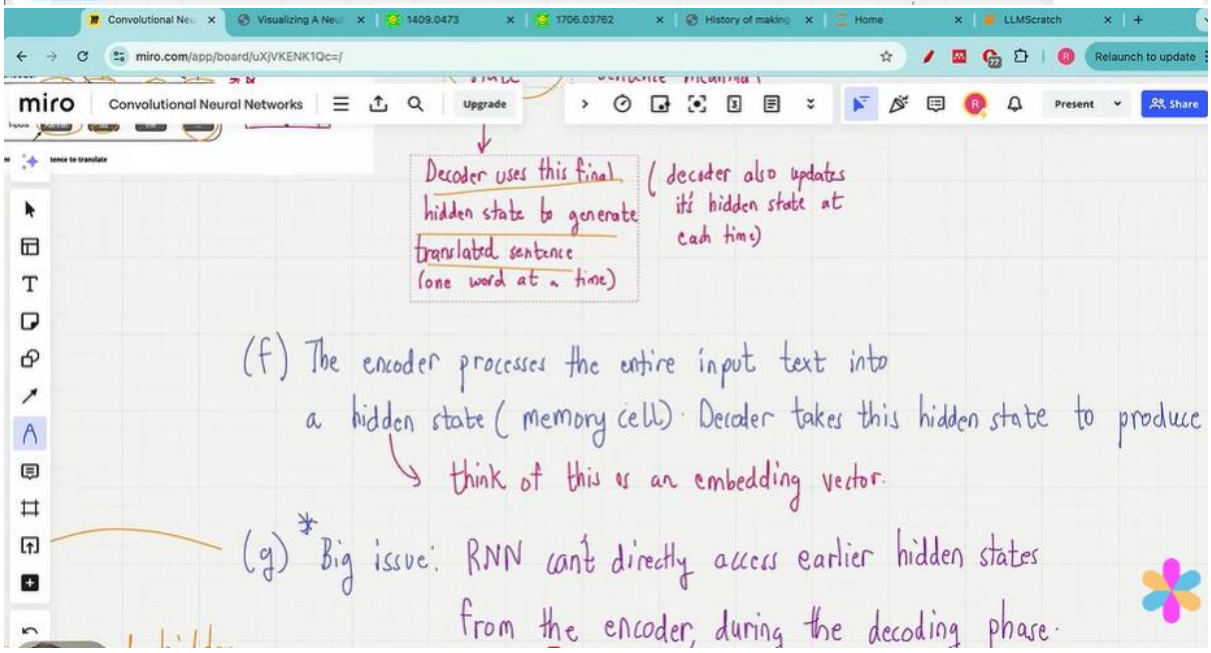

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Encoder
Decoder
Reads and processes text
Translates text
(c) Before transformers, Recurrent Neural Networks (RNN
were the most popular encoder-decoder architecture
for language translation.

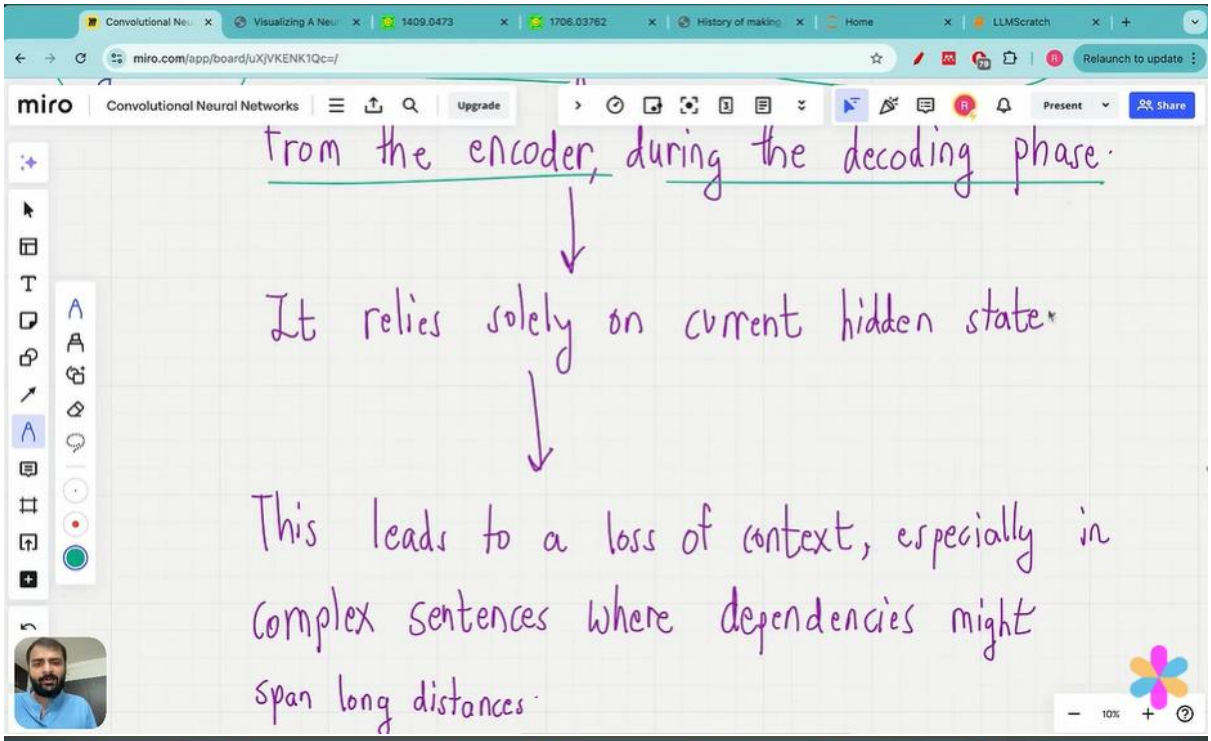

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(d) RNN: Output from previous step is fed as input to
current text
(e) Here's how the encoder-decoder RNN works:
Input text
Encoder (processes input text
sequentially)
Updates hidden (internal values

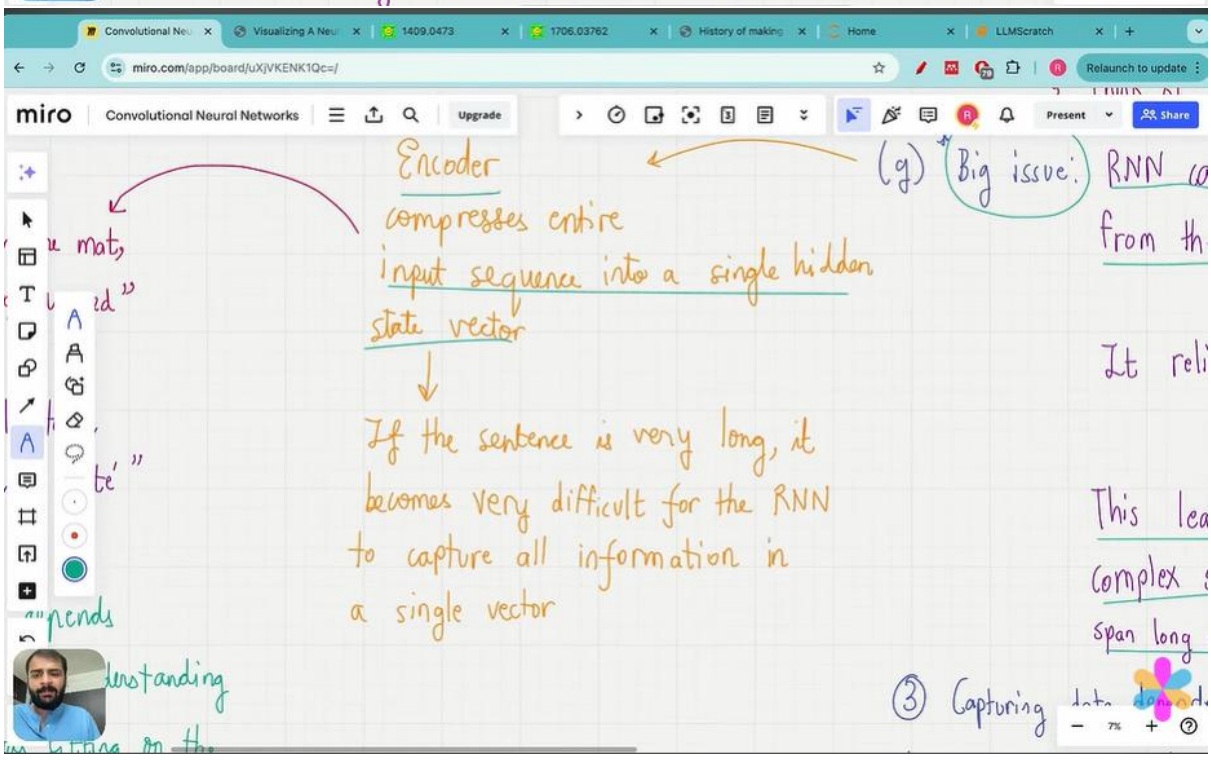

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Input text
Encoder (processes input text sequentially)
Updates hidden state at each step (internal values at hidden layers)
12%

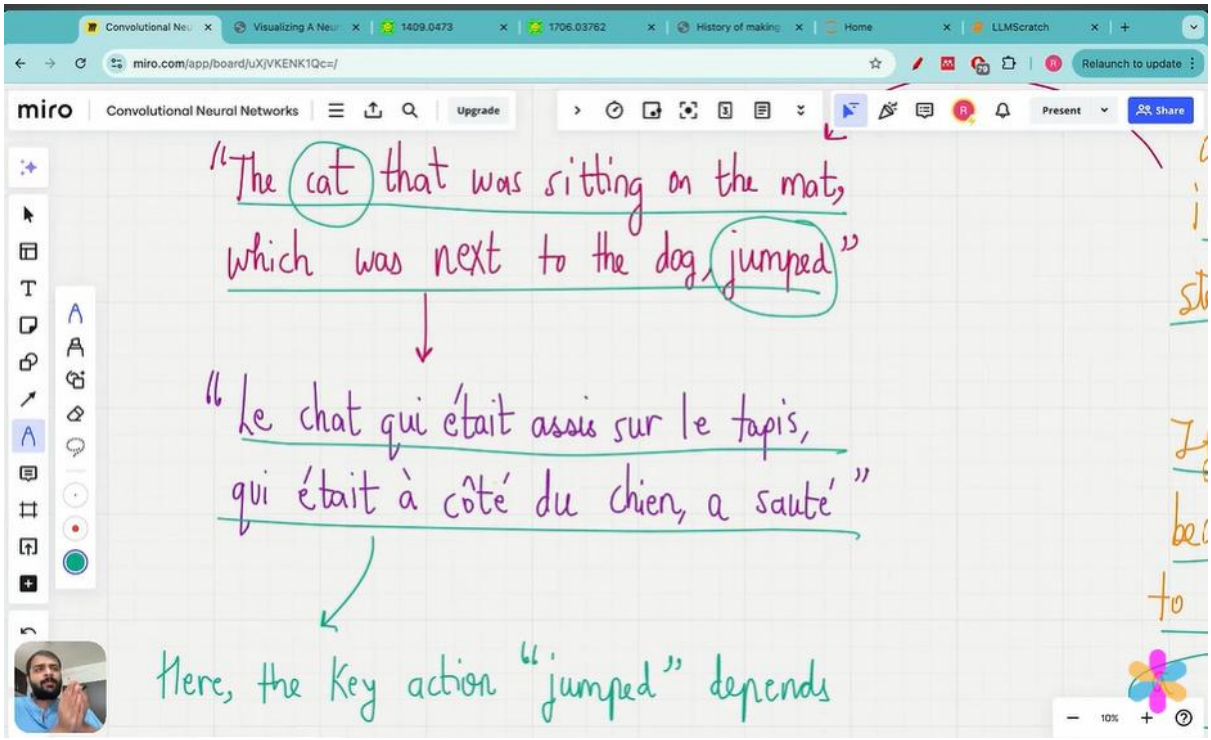

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The translated English sentence
Can
you
help
...
A memory cell (hidden state) memorizing entire input
Final hidden state (encoder tries to capture sentence meaning)
Decoder uses this final hidden state to generate translated sentence (one word at a time)
(decoder also updates it's hidden state at each time)
12%

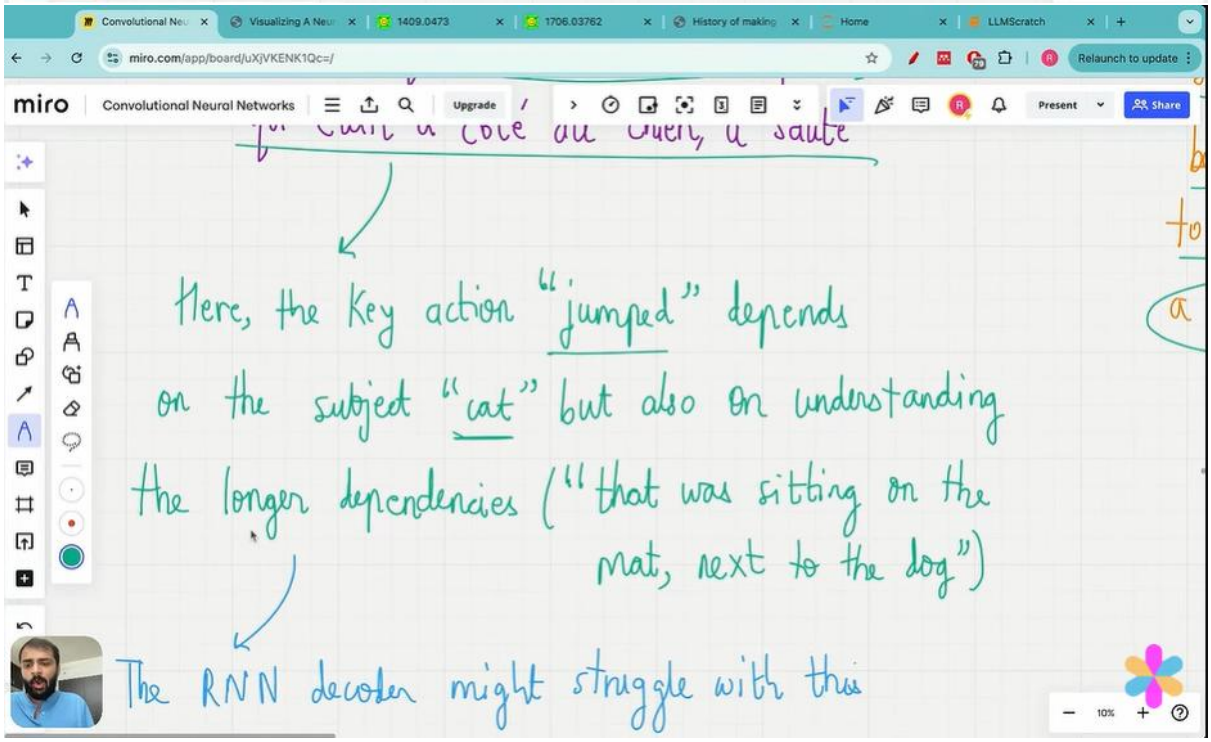

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Decoder
The translated English sentence
Outputs
Can
you
help
...
Hidden states
Hidden states of a neural network
Encoder
Inputs
Kannst
du
mir
...
A memory cell (hidden state) memorizing entire input
German input sentence to translate
step
Final hidden state
Decoder uses this
hidden state to g
translated sentence
(one wor

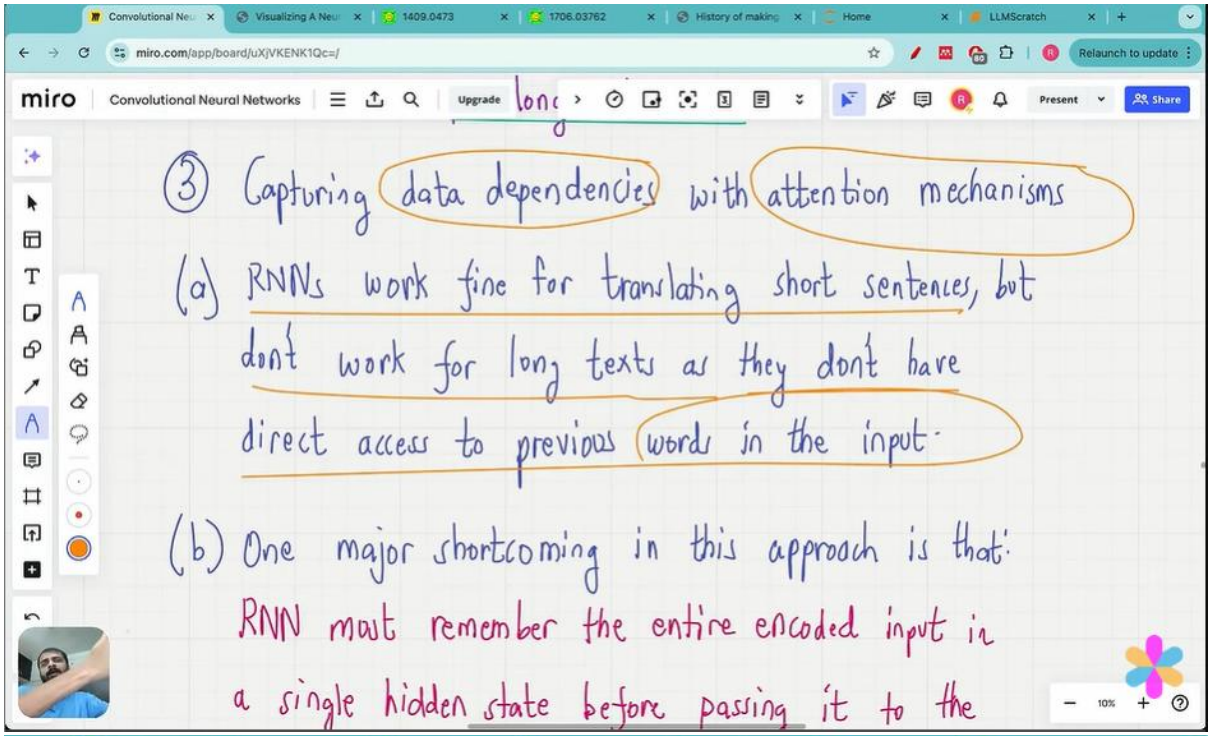

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Decoder uses this final hidden state to generate translated sentence (one word at a time)
(decoder also updates its hidden state at each time)
(f) The encoder processes the entire input text into a hidden state (memory cell). Decoder takes this hidden state to produce
think of this as an embedding vector.
(g) * Big issue: RNN can't directly access earlier hidden states from the encoder, during the decoding phase.

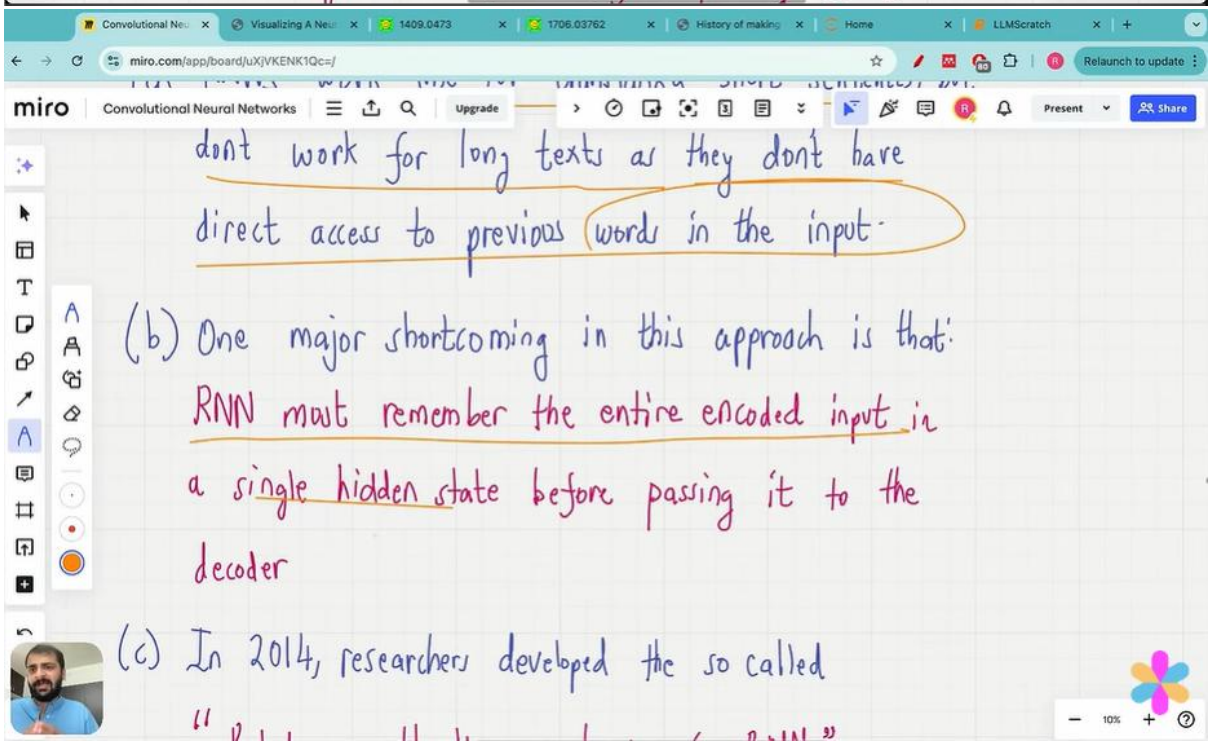

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from the encoder, during the decoding phase.
It relies solely on current hidden state.
This leads to a loss of context, especially in complex sentences where dependencies might span long distances.
10%

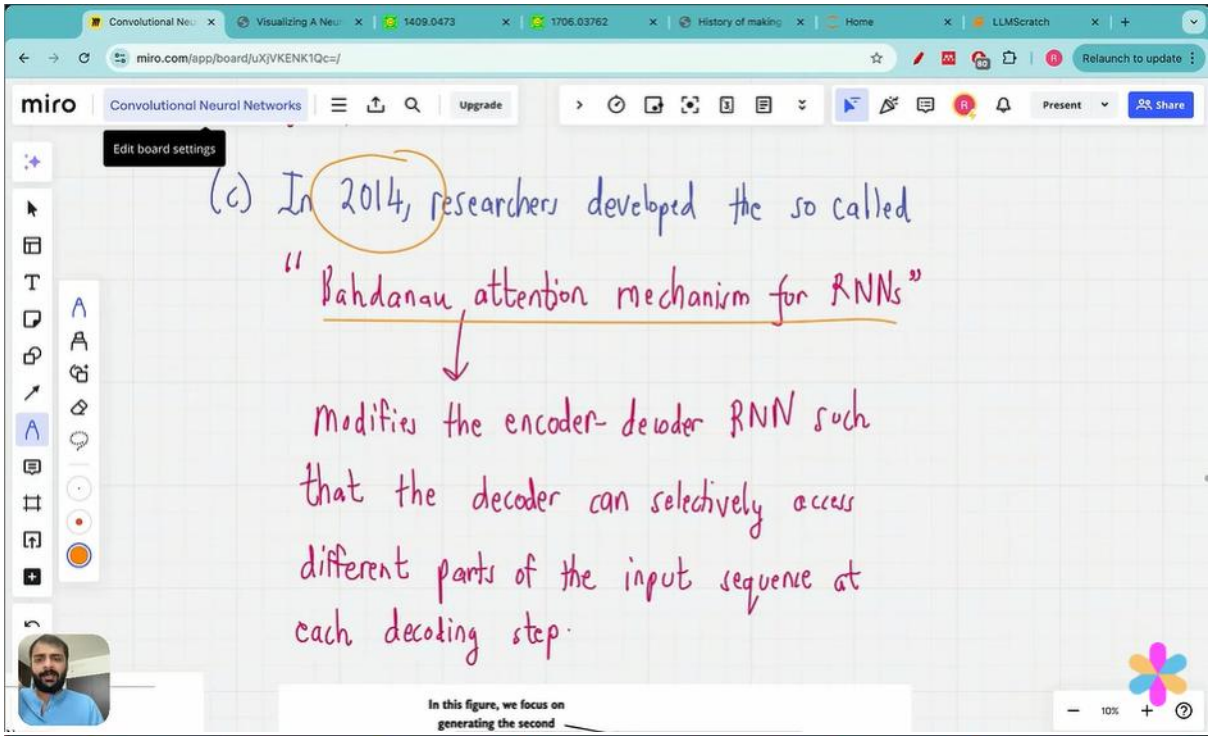

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Encoder compresses entire input sequence into a single hidden state vector
If the sentence is very long, it becomes very difficult for the RNN to capture all information in a single vector
(g) Big issue:
RNN
from th
It reli
This lea
complex
span long
③ Capturing
7%

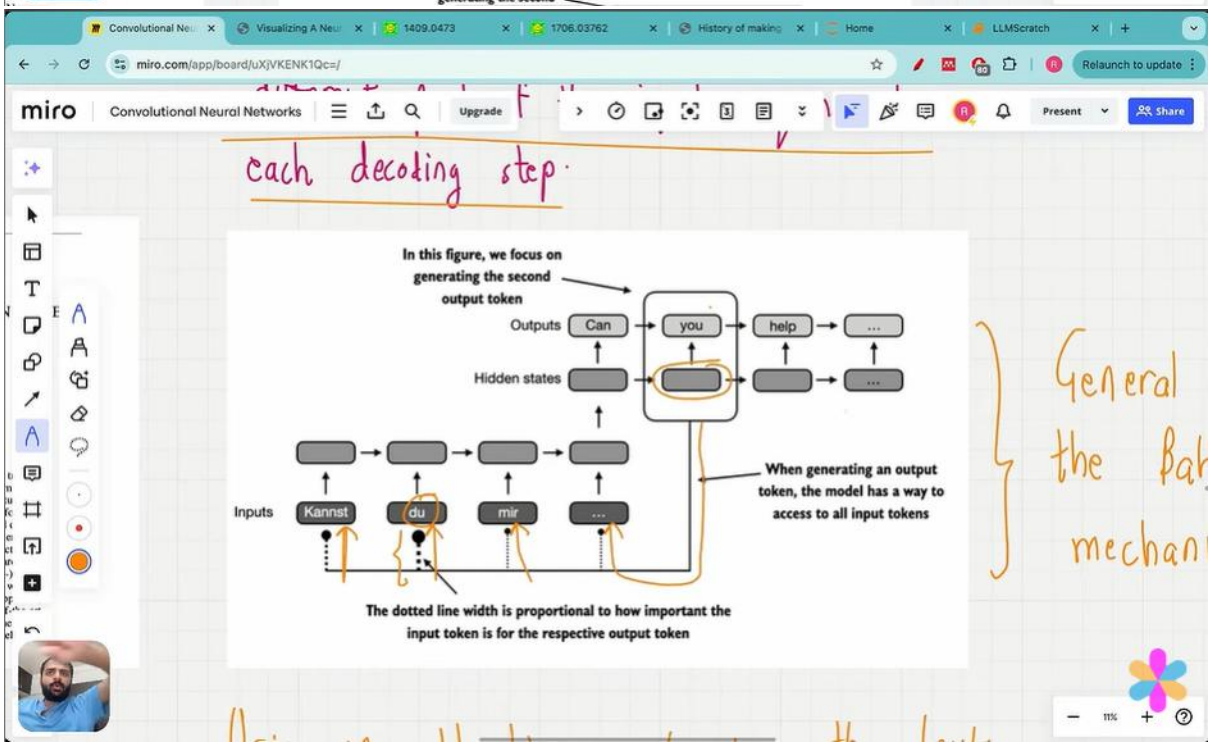

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"The cat that was sitting on the mat, which was next to the dog, jumped"
" Le chat qui était assis sur le tapis, qui était à côté du chien, a sauté "
Here, the Key action "jumped" depends

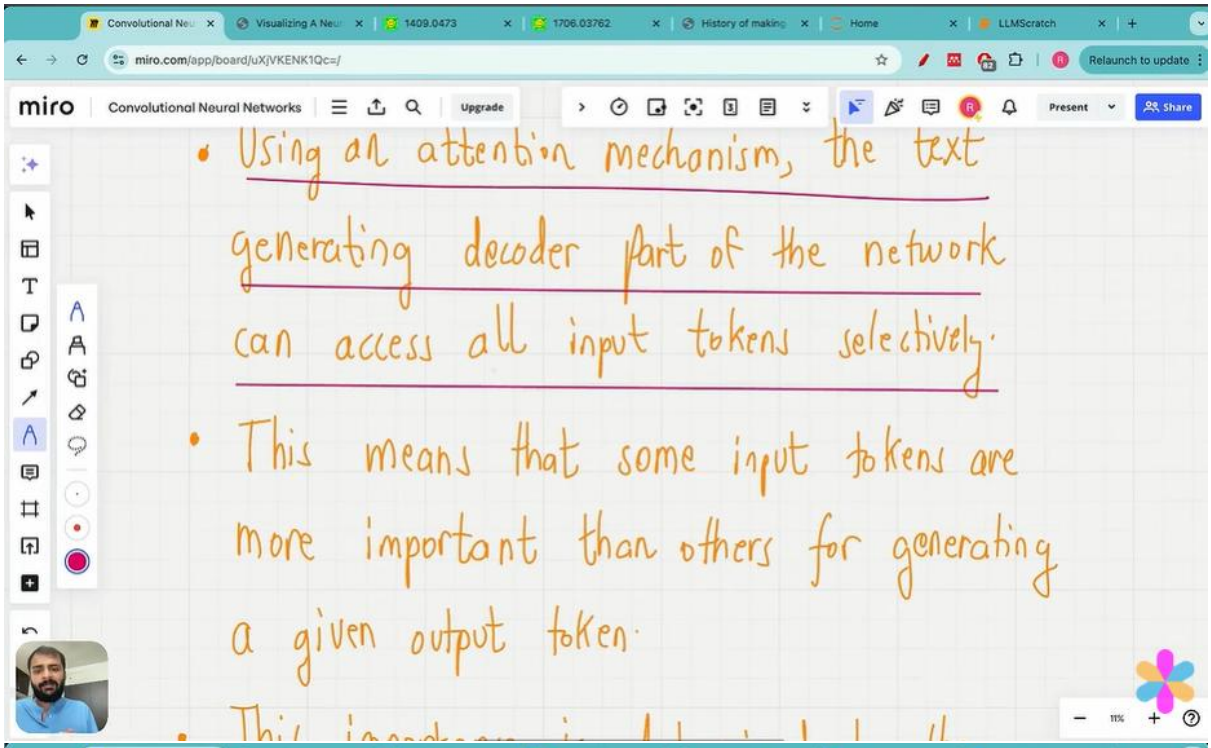

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Here, the Key action "jumped" depends
on the subject "cat" but also on understanding
the longer dependencies ("that was sitting on the
mat, next to the dog")
The RNN decoder might struggle with this

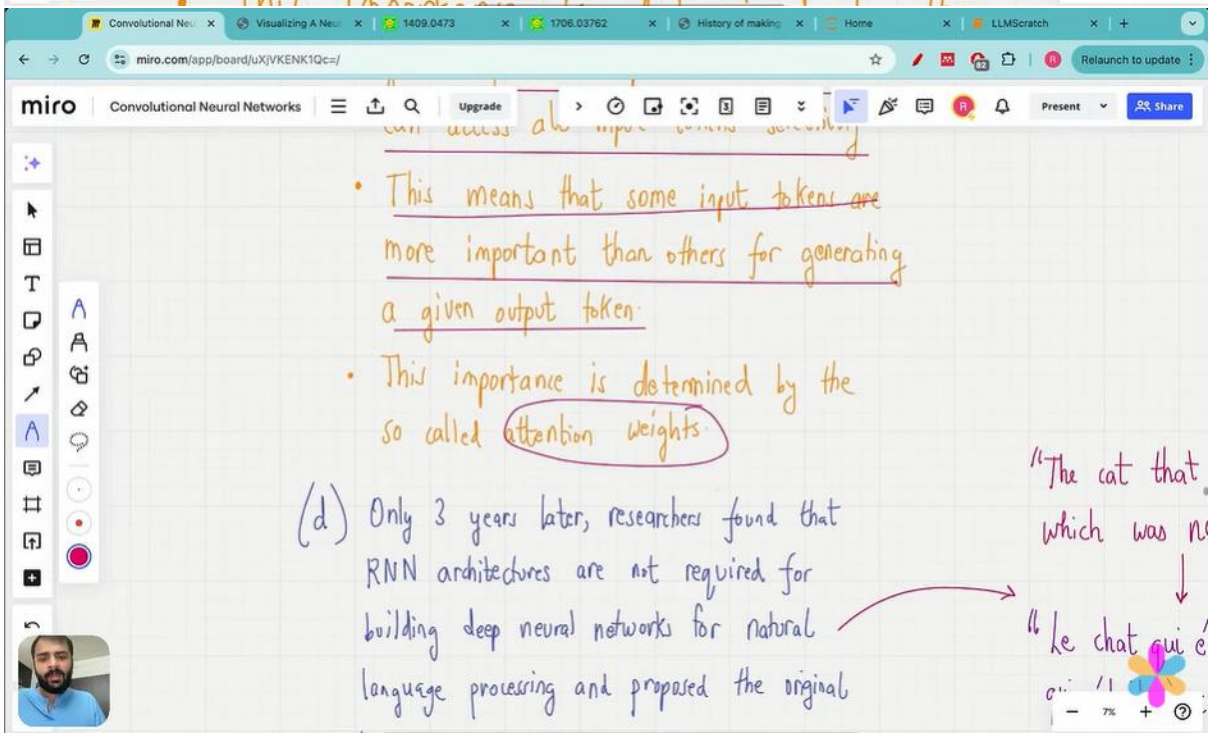

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③ Capturing data dependencies with attention mechanisms
(a) RNNs work fine for translating short sentences, but dont work for long texts as they dont have direct access to previous words in the input.
(b) One major shortcoming in this approach is that: RNN must remember the entire encoded input in a single hidden state before passing it to the
10%

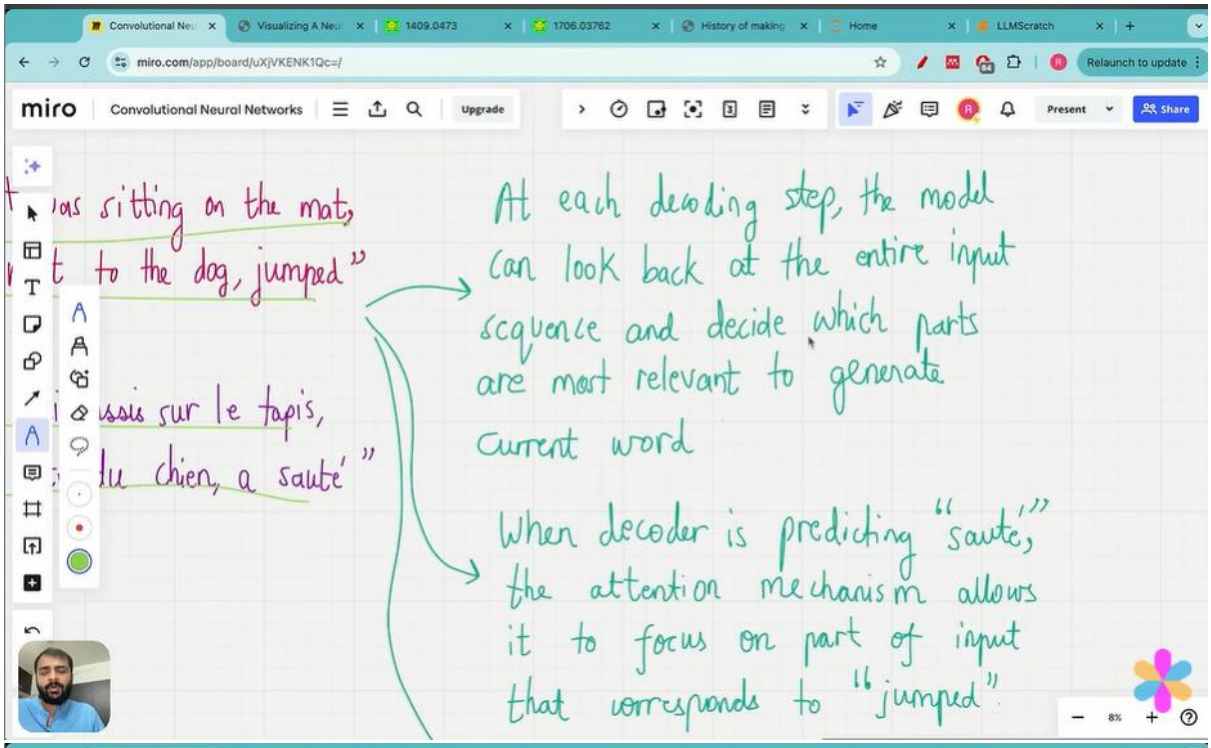

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dont work for long texts as they dont have direct access to previous words in the input.
(b) One major shortcoming in this approach is that: RNN must remember the entire encoded input in a single hidden state before passing it to the decoder
(c) In 2014, researchers developed the so called
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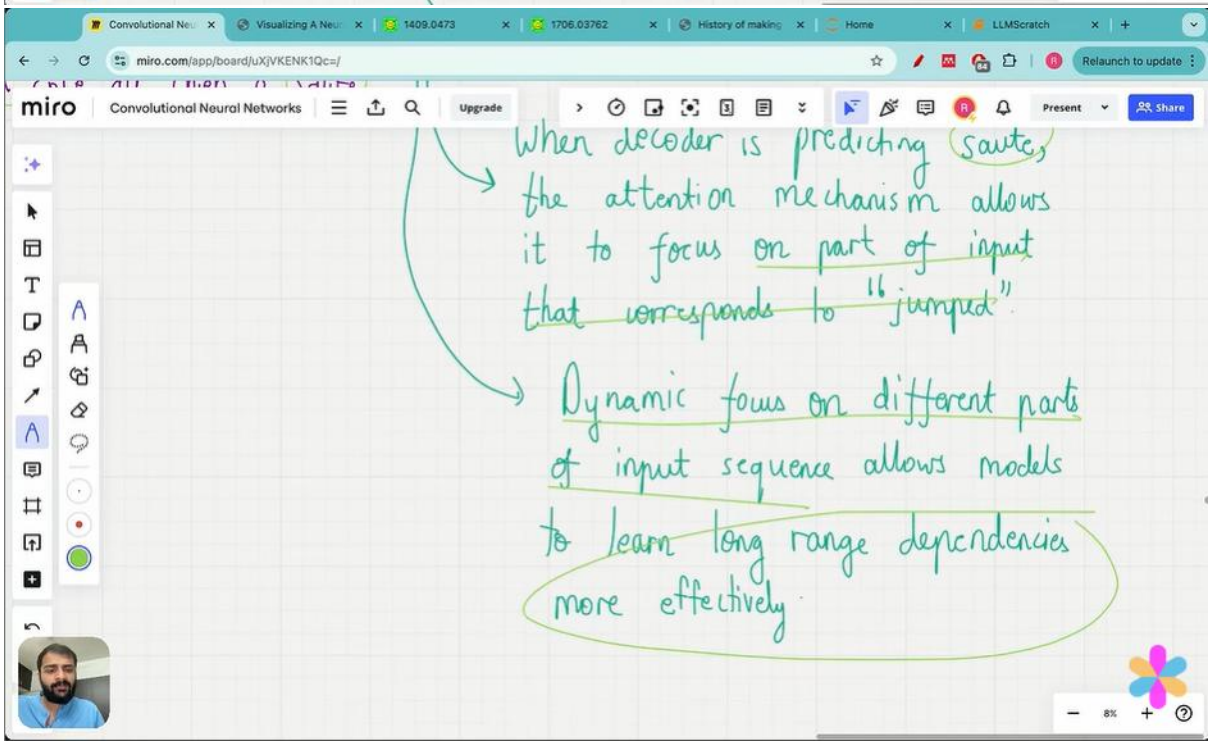

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(c) In 2014, researchers developed the so called
" Bahdanau attention mechanism for RNNs"
modifies the encoder-decoder RNN such that the decoder can selectively access different parts of the input sequence at each decoding step.
In this figure, we focus on generating the second

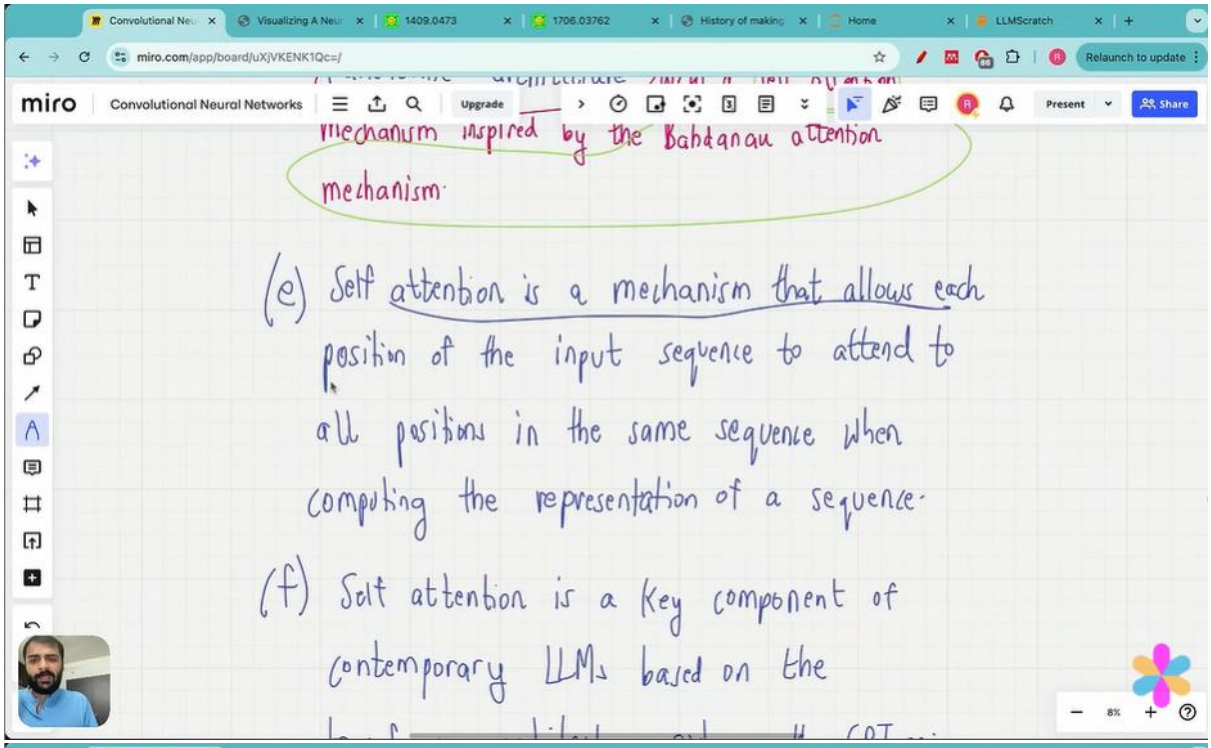

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each decoding step.
In this figure, we focus on generating the second output token
Outputs
Can
you
help
...
Hidden states
Inputs
Kannst
du
mir
...
When generating an output token, the model has a way to access to all input tokens
The dotted line width is proportional to how important the input token is for the respective output token
General
the Bah
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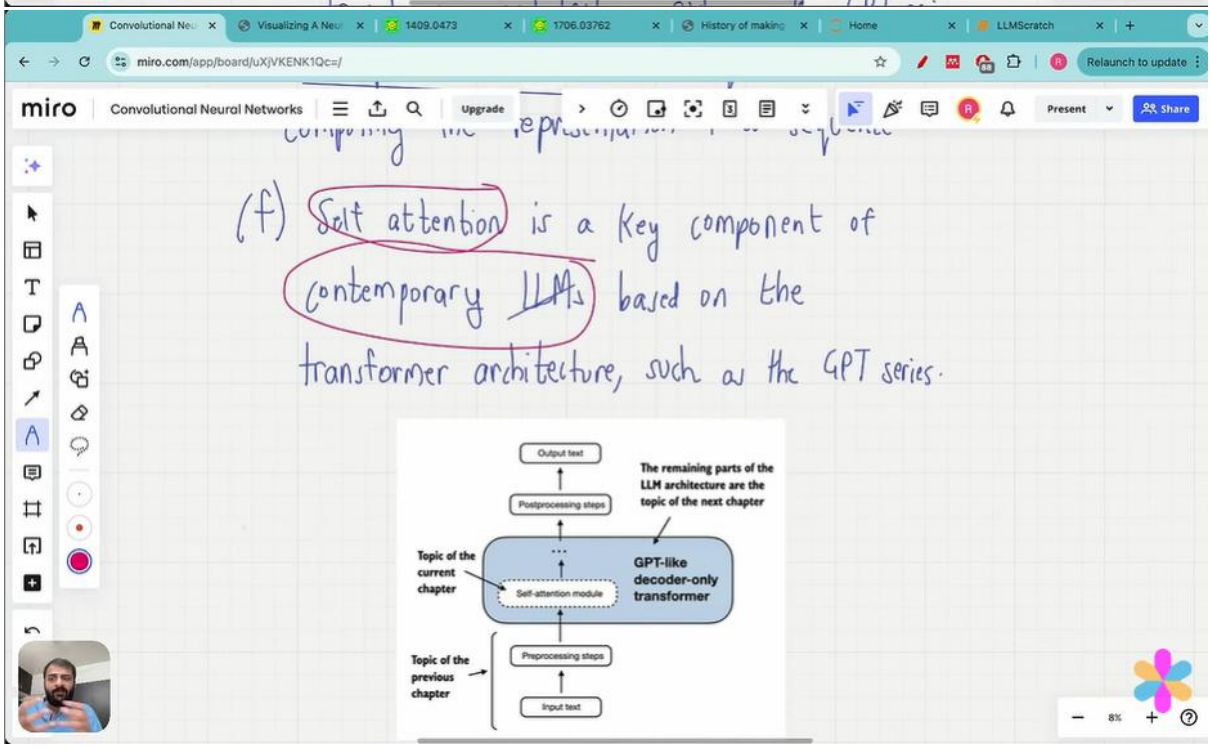

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• Using an attention mechanism, the text generating decoder part of the network can access all input tokens selectively.
• This means that some input tokens are more important than others for generating a given output token.

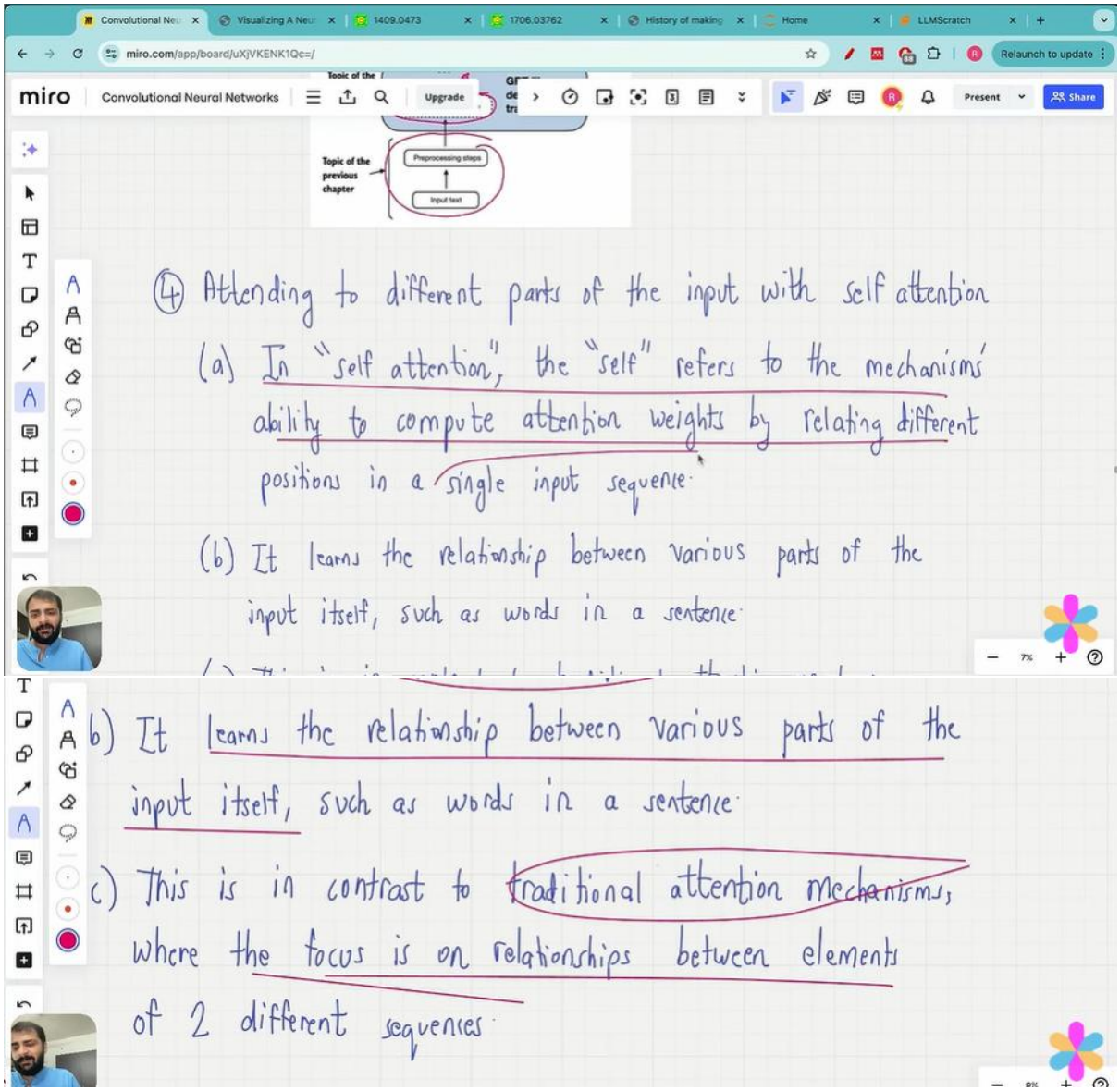

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• This means that some input tokens are more important than others for generating a given output token.
• This importance is determined by the so called attention weights.
(d) Only 3 years later, researchers found that RNN architectures are not required for building deep neural networks for natural language processing and proposed the orginal
"The cat that
which was


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as sitting on the mat,
t to the dog, jumped"
ssis sur le tapis,
le chien, a sauté"
At each decoding step, the model can look back at the entire input sequence and decide which parts are most relevant to generate current word
When decoder is predicting "saute,"
the attention mechanism allows it to focus on part of input that corresponds to "jumped".
8%


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When decoder is predicting "saute,"
the attention mechanism allows it to focus on part of input that corresponds to "jumped".
Dynamic focus on different parts of input sequence allows models to learn long range dependencies more effectively.
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mechanism inspired by the Bahdanau attention mechanism.
(e) Self attention is a mechanism that allows each position of the input sequence to attend to all positions in the same sequence when computing the representation of a sequence.
(f) Self attention is a key component of contemporary LLMs based on the
8%


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(f) Self attention is a key component of contemporary LLMs based on the transformer architecture, such as the GPT series.
Output text
Postprocessing steps
The remaining parts of the LLM architecture are the topic of the next chapter
Topic of the current chapter
GPT-like decoder-only transformer
Self-attention module
Topic of the previous chapter
Preprocessing steps
Input text
8%


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Topic of the previous chapter
Preprocessing steps
Input text
④ Attending to different parts of the input with self attention
(a) In "self attention", the "self" refers to the mechanisms' ability to compute attention weights by relating different positions in a single input sequence.
(b) It learns the relationship between various parts of the input itself, such as words in a sentence.
7%
(b) It learns the relationship between various parts of the input itself, such as words in a sentence.
(c) This is in contrast to traditional attention mechanisms, where the focus is on relationships between elements of 2 different sequences.